

SARUPYA MITRA

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	M.Tech (Vision and Intelligent Systems)	IIT Kharagpur	9.18 / 10
2021	B.Tech (Electronics and Communication Engineering)	University of Engineering and Management, Kolkata	9.57 / 10
2017	Higher Secondary	West Bengal Council of Higher Secondary Education	82%
2015	Secondary	West Bengal Board of Secondary Education	84.28%

PROJECTS

Facial Expression Recognition using Hand-crafted Features

[Documented Repository](#)

- Implemented a **multi-scale, multi-orientation Gabor filter bank from scratch** for facial feature extraction.
- Extracted **block-wise orientation energy statistics** from grayscale facial images in the **CK+ dataset**.
- Trained a **Linear SVM** classifier for multi-class emotion recognition and evaluated the impact of different Gabor scales and orientations on classification performance.
- Achieved a best classification accuracy of **80.1%** using two Gabor scales and eight orientations.
- Observed diminishing performance gains with additional scales due to feature redundancy and dataset size constraints.

Underwater Image Enhancement using Classical Computer Vision Techniques

[Documented Repository](#)

- Hand-coded **four underwater image enhancement algorithms**, including **Max-RGB Automatic White Balancing + CLAHE** and **multiple Dark Channel Prior(DCP) based restoration models**.
- Designed a **red channel compensation preprocessing technique** using chromatic normalization to mitigate wavelength-dependent color attenuation in underwater environments.
- Implemented a **modified Dark Channel Prior** variant using **minimum of green-blue channels** to reduce bias caused by red channel absorption.
- Evaluated models on the **UIEB(Underwater Image Enhancement Benchmark) dataset** using **PSNR, SSIM and UIQM** metrics; all models demonstrated improved perceptual underwater image quality, with the AWB+CLAHE model achieving the highest mean UIQM score of **~ 2.19**.

Skin Lesion Classification using SimCLR

[Documented Repository](#)

- Built the full **SimCLR self-supervised learning pipeline** for multi-class skin lesion classification using the DermaMNIST dataset.
- Implemented **contrastive learning (NT-Xent loss) with custom augmentations** to learn visual representations without labeled data.
- Evaluated **downstream performance via linear classifier**, achieving **70% accuracy** on an imbalanced dataset.
- Performed error analysis using **confusion matrix and classification metrics**, identifying bias towards the majority class and revealing poor minority class recall.
- Visualized embedding space using **t-SNE**, revealing poor class separability and overlap between majority and minority classes.

Image Classification using Vision Transformer(ViT)

[Documented Repository](#)

- Implemented a **Vision Transformer (ViT)** independently in PyTorch (without pre-built transformer modules), training on **CIFAR-100 and Food-101** datasets.
- Performed **ablation studies** across patch size, embedding dimension, transformer depth, dropout, and LR scheduling, improving CIFAR-100 test accuracy from **~42% to ~53%**.
- Observed improved ViT performance with **increased data scale**, achieving **~61.5% accuracy on Food-101** vs **~53% on CIFAR-100** under similar regularization.
- Identified **dropout after the MLP layer** as the most impactful regularization technique, improving CIFAR-100 test accuracy by **~6%** and reducing overfitting.

WORK EXPERIENCE

Jr. Software Engineer | Cognizant

Jul 2021 - Oct 2023

- Enhanced **application security** for a client-facing.NET application by migrating password hashing from **MD5** to **bcrypt**, ensuring compliance with **modern cryptographic standards** and eliminating a critical vulnerability in the authentication layer.
- Independently diagnosed and resolved production-level defects in **.NET Framework applications** by directly identifying root causes and modifying application code, improving reliability and user satisfaction.
- Maintained **structured documentation** of bug resolution workflows and root causes analyses across incident categories, enabling **faster diagnosis** of repeated defect patterns.

POSITIONS OF RESPONSIBILITY

Teaching Assistant | IIT Kharagpur

Jul 2025 - Apr 2026

- Mentored and evaluated **72 students** across two advanced laboratory courses.
- Conducted lab sessions, **clarified conceptual doubts** and guided students through implementation of image processing and ML algorithms.
- Assessed lab submissions and provided structured feedback, supporting consistent student performance across the semester.

SKILLS AND EXPERTISE

Programming Languages: Python, SQL | **Libraries:** Numpy, Matplotlib, Pytorch, OpenCV

AWARDS AND ACHIEVEMENTS

- Secured All India Rank **359** in GATE ECE 2024
- Awarded **Chancellor's Award for Academic Excellence** by University of Engineering and Management, Kolkata in 2018 and 2021 for securing **1st** position in the ECE Department.